

Week 5: Potential Energy and Energy Conservation – Supplementary Notes

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These notes support the Week 5 lecture. They contain background that was moved out of the contact time and are not needed for the lecture problems. Sections marked *extension* are not assessed.

1 A derivative refresher

$\frac{d}{dx}x^n = nx^{n-1}$ for any n (including negative n , for example $\frac{d}{dx}x^{-2} = -2x^{-3}$). For constants, $\frac{d}{dx}(cf) = c\frac{df}{dx}$, and $\frac{d}{dx}(f+g) = \frac{df}{dx} + \frac{dg}{dx}$. Example: $y = 3x^2 - x^3 \Rightarrow y' = 6x - 3x^2$. The derivative is the slope of the tangent to the graph. A point where $y' = 0$ is a maximum, a minimum or a point of inflexion; the sign of the second derivative y'' decides which (Section 6).

2 Why the work around a closed loop is zero

Suppose a force does the same work on every path between two points A and B (path independence). Take any closed loop and split it at two points A and B into path 1 (A to B) and path 2 (B back to A). The work around the loop is $W_{\text{loop}} = W_1 + W_2$. Reversing the direction of travel along a path reverses the sign of $\vec{F} \cdot d\vec{r}$ at every step, so the work from B to A along path 2 is the negative of the work from A to B along path 2. By path independence, A to B along path 2 equals A to B along path 1. Therefore $W_{\text{loop}} = W_1 - W_1 = 0$. Conversely, if the work around every closed loop is zero, the two paths between any A and B must give the same work. The two statements are equivalent. Kinetic friction fails the test because its work is negative on every segment, in both directions.

3 Where does $F_x = -dU/dx$ come from?

For a conservative force, $W = -\Delta U$. For a very small displacement dx along x , the force is effectively constant, so $F_x dx = -dU$. Dividing by dx gives $F_x = -\frac{dU}{dx}$. Examples: $U = \frac{1}{2}kx^2 \Rightarrow F_x = -kx$; $U = mgy \Rightarrow F_y = -mg$; and for the potential energy in Week 5, $U = 6.0x^2 - 2.0x^3 \Rightarrow F_x = -12x + 6x^2$.

4 Extension: forces in three dimensions

In three dimensions, $U(x, y, z)$ gives $F_x = -\partial U/\partial x$, $F_y = -\partial U/\partial y$, $F_z = -\partial U/\partial z$, or $\vec{F} = -\nabla U$. Each partial derivative treats the other two coordinates as constants. Example: $U = mgz$ gives $\vec{F} = -mg\hat{k}$. This is not needed for the lecture problems.

5 The vertical loop

A block of mass m moves round the inside of a smooth vertical loop of radius R . At the top, the forces toward the centre are the weight mg and the normal force N : $N + mg = mv_{\text{top}}^2/R$. The block stays in contact only if $N \geq 0$, that is $v_{\text{top}}^2 \geq gR$. Energy conservation from the bottom to the top gives $v_{\text{bottom}}^2 = v_{\text{top}}^2 + 4gR$, so $v_{\text{bottom}}^2 \geq 5gR$. If the block starts from rest at height h above the bottom of the loop, then $h \geq 2.5R$. The same argument holds for a stone on a string, with N replaced by the tension T .

6 Stability of an equilibrium point

Equilibrium occurs where $dU/dx = 0$. If $d^2U/dx^2 > 0$ (a minimum of U), a small displacement gives a restoring force: the equilibrium is *stable*. If $d^2U/dx^2 < 0$ (a maximum), a small displacement gives a force away from the point: *unstable*. For $U = 6.0x^2 - 2.0x^3$: $U'' = 12 - 12x$, which is $+12$ at $x = 0$ (stable) and -12 at $x = 2.0$ m (unstable). Near a stable minimum, $U \approx U_0 + \frac{1}{2}U''(x - x_0)^2$, a spring-like potential, which is why small oscillations are described by Hooke's law (later in the course).

7 Preview: gravitational potential energy over large distances

Near the surface of the Earth, $U = mgy$. For large distances from a mass M , the force is $F = -GMm/r^2$ (attractive, toward the centre), and $F_r = -dU/dr$ with $U \rightarrow 0$ as $r \rightarrow \infty$ gives

$$U(r) = -\frac{GMm}{r}.$$

U is negative and increases toward zero as r increases. The Gravitation lecture uses this to find the escape speed from $\frac{1}{2}mv^2 - GMm/R = 0$. Near the surface, $U(R + y) - U(R) \approx mgy$ with $g = GM/R^2$.

8 Energy dissipation and temperature

When kinetic friction acts, the mechanical energy lost, $f_k d$, appears as an increase ΔE_{int} in the internal (thermal) energy of the object and the surface. Part of the energy flows to the surroundings as heat. How internal energy relates to temperature is the subject of the thermal physics part of the course (the first law of thermodynamics). For the problems in this lecture, treat $f_k d$ as the energy that has left the mechanical energy bookkeeping.

9 Units and conversions

1 J = 1 N m = 1 kg m²/s²; 1 eV = 1.60 × 10⁻¹⁹ J; 1 kWh = 3.6 × 10⁶ J; 1 cal = 4.19 J (the food Calorie is 1000 calories). A fall of 1 m changes the gravitational potential energy of a 1 kg mass by 9.8 J; a 1 kg mass moving at 1 m/s has kinetic energy 0.5 J.